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Title: Impaired efferocytosis by synovial macrophages in patients with knee osteoarthritis

Running Title: Impaired efferocytosis in knee osteoarthritis

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Abstract

Objective: Osteoarthritis (OA) exposes all joint tissues to physiologic stresses, increasing the need to clear apoptotic cells from tissues including the synovium. Our objectives were to assess the burden of apoptotic cells in synovial tissue in patients with late-stage knee OA, and to investigate whether OA impairs the macrophage-mediated clearance of apoptotic cells via efferocytosis.

Methods: Synovial tissues were collected from individuals with healthy knees and patients with late-stage knee OA during arthroplasty. Synovial apoptotic cell burden was assessed by immunofluorescence for cleaved caspase-3. Efferocytosis of apoptotic Jurkat cells by CD14+ synovial tissue macrophages, and peripheral blood-derived macrophages was quantified using immunofluorescence microscopy. Effects of OA on macrophage-mediated efferocytosis were modeled by stimulating blood-derived macrophages with synovial fluid collected from individuals with healthy knees, early-, or late-stage knee OA.

Results: Patients with late-stage knee OA had more apoptotic synovial cells compared with healthy synovium. There was a marked reduction in the fraction of synovial tissue macrophages engaging in efferocytosis and quantity of material efferocytosed by individual macrophages. Blood-derived macrophages exposed to synovial fluid from patients with knee OA recapitulated the defective efferocytosis, with the greatest effect from patients with early-stage knee OA and higher pain and inflammation.

Conclusion: Apoptotic cells accumulate in the synovium of patients with late-stage knee OA. Our results suggest that OA impairs critical homeostatic functions of synovial macrophages leading to accumulation of apoptotic cells.

Introduction

Synovial lining (synovium) function is critical for maintaining joint health. Synovial tissue dysfunction may therefore worsen joint outcomes in osteoarthritis (OA). As in most tissues, macrophages play a vital role in maintaining synovial tissue homeostasis. In healthy synovium, resident macrophages clear debris and recycle the lubricating components of synovial fluid (1). Macrophages also remove apoptotic cells via efferocytosis, an immunologically silent process with pro-resolving effects under inflammatory conditions (2). Efferocytosis is required to avoid post-apoptotic secondary necrosis that can promote inflammation and disruption of homeostasis (3). Since the accumulation of apoptotic cells could lead to synovial tissue dysfunction, we examined the presence of apoptotic cells in the synovium of patients with late-stage knee OA and investigated the effects of OA on synovial macrophage-mediated clearance of dead and dying cells via efferocytosis.

The synovium consists of an intima and subintima and supports the metabolic demands of all joint tissues, replenishes synovial fluid with nutrients and lubricating molecules, and clears debris from normal turnover of joint tissues (4,5). OA-related synovial inflammation (synovitis) manifests as synovial lining thickening (hyperplasia and hypertrophy), leukocyte infiltration from peripheral blood, and effusion (6,7). Ultrasound (US) and magnetic resonance imaging (MRI) have demonstrated that synovitis severity is associated with worse pain and disease progression of OA (8,9). Recruited synovial macrophages release pro-inflammatory cytokines and enzymes in the setting of chronic inflammation and may exacerbate OA progression (10), but macrophages are also important to inflammation resolution (11,12). Thus, failure of macrophages to successfully resolve chronic synovial inflammation during OA may contribute to

poor joint health outcomes. However, the mechanisms underlying failed inflammation resolution in OA are not well-understood.

Apoptotic cells accumulate in tissues in multiple diseases. For example, dysregulation of efferocytosis in atherosclerosis leads to a buildup of apoptotic cells in the atherosclerotic plaques within the vasculature, which further contributes to the progression and severity of the disease (13). In OA, it is well-established that apoptotic chondrocytes contribute to OA pathogenesis (14,15), but there is virtually no information available about apoptotic synovial cells in OA. Uncleared apoptotic cells could contribute to chronic inflammation in chronic joint diseases including OA. However, no studies have analyzed whether apoptotic cells accumulate in the synovium of OA knee joints, nor whether the joint microenvironment influences efferocytosis. Our objectives were 1) to investigate if apoptotic cells accumulate within the synovial lining of patients with late-stage knee OA and 2) to test whether OA is associated with impaired macrophage-mediated efferocytosis.

Materials and Methods

Materials

Cell culture and efferocytosis reagents were from Fisher Scientific and Sigma-Aldrich, and tissue culture plastics from BD Falcon.

Study design and participants

Research involving patients and healthy volunteers was conducted with written, informed consent and approved by the Health Sciences Research Ethics Board at the University of Western Ontario (REB #109255). Tissue, blood, and synovial fluid were collected from patients

with early- and late-stage knee OA and volunteers with healthy knees enrolled in the Western Ontario Registry for Early Osteoarthritis (WOREO) Knee Study, a prospective cohort designed to investigate roles of inflammation and biomechanics in knee OA. Additional fresh, healthy (non-OA) synovial tissues were collected from cadaveric donors with approval from the University of Western Ontario Subcommittee for Cadaveric Material Research Ethics (#08072019). All patients and volunteers with healthy knees underwent a clinical assessment to diagnose or rule out symptomatic knee OA, respectively, by a rheumatologist or orthopaedic surgeon. All participants completed questionnaires for frequent knee symptoms ('Yes'/'No' to any pain, aching, or stiffness in each knee on most days for >4 weeks currently, and at any point in their life) and the knee-specific Knee Injury and Osteoarthritis Outcome Score (KOOS) pain subscale score. Knee radiographs and musculoskeletal US of the knee following the OMERACT US protocol (16) were also collected.

Synovial fluid collection

Synovial fluid samples were collected under direct US-guidance or by aspiration immediately prior to arthrotomy at the time of arthroplasty. We excluded synovial fluid samples visibly contaminated by blood. Synovial fluid samples were placed immediately on ice, centrifuged within 2 hours at 2800 x g at 4°C for 2 x 10-minute cycles to remove cells and debris, and stored at -80°C prior to use.

Synovial tissue collection

Healthy volunteers

Healthy synovial tissues were from volunteers with healthy knees. Absence of knee pathology was confirmed in volunteers by i) no frequent knee symptoms, ii) normal knee joint radiographs (Kellgren-Lawrence; KL grade 0; 17), iii) musculoskeletal US of the knee demonstrating no sign of synovitis or cartilage pathology (OMERACT knee synovitis and cartilage grades 0; 16), and iv) absence of moderate to severe synovitis scores by synovial histopathology. Using a standard lateral approach (18), US-guided needle biopsy with a 16 ga. x 19 mm side-cut needle biopsy (Argon Medical, USA) was used to acquire 3-5 cores of synovial tissue from volunteers with healthy knees.

Healthy human cadavers

Healthy synovial tissues were collected by lateral arthrotomy from cadaveric donors within 24 hours of death. Absence of knee pathology in cadaveric donors was confirmed by i) report from family members at the time of death that the donor did not complain of frequent knee joint symptoms, ii) absence of synovitis on musculoskeletal US of the knee at the time of tissue collection, iii) visual confirmation of no or minimal cartilage pathology on open knee examination of the femoral condyles and trochlea (Collins grade 0-I; 19), and iv) absence of moderate to severe infiltrate grades by synovial histopathology.

Total knee arthroplasty patients

Synovial tissue samples were collected from the lateral suprapatellar recess of the knee using a standard parapatellar approach (18) at the time of total knee arthroplasty. Samples were obtained from the lateral suprapatellar synovial recess to standardize anatomical variability, avoid synovial fat pads, align with US assessment, and focus on a joint region where synovitis typically occurs in knee OA.

Histopathology and immunofluorescence

Synovial tissue biopsies were placed in 4% paraformaldehyde (PFA; Sigma-Aldrich, P6148-1KG) for 16 hours, processed for paraffin embedding, and sectioned at 5 μ m. Slides were stained with Harris Hematoxylin and 0.25% Eosin Y (H&E), mounted with Cytoseal XLY (Richard-Allan Scientific), and graded by a blinded rater on six parameters: 1) synovial lining thickness, 2) sub-synovial infiltration by leukocytes, 3) surface fibrin deposition, 4) vascularization, 5) fibrosis, and 6) perivascular edema, on a 0–3 scale (0 = normal, 1 = mild, 2 = moderate, 3 = severe) (20). Grading was completed on 5 slides per patient and mean scores were assigned into categories 0, 0.5–1.5, and 2–3 (none, mild, moderate-severe).

Antigen retrieval for immunofluorescence was performed with 70°C Tris buffer (10 mM Tris; 1 mM EDTA; 10% glycerol). Blocking and permeabilization was achieved with 0.2% Triton X (Sigma-Aldrich; 9002-93-1) and 5% bovine serum albumin (BSA; GE Healthcare) in 1X phosphate buffered saline (PBS) solution, respectively. Primary antibody (Abcam) incubation was performed overnight at 4°C for apoptotic cells with rabbit anti-human cleaved caspase-3 (CC3; 1:1000, ab49822), and synovial macrophages with rabbit anti-human CD14 (1:200, ab183322) or mouse anti-human CD68 (1:200, ab201340). Negative controls consisted of Rabbit IgG isotype primary antibody (Abcam, ab171870) on adjacent sections. Secondary antibodies (Jackson ImmunoResearch) were incubated for 1 hour at room temperature in the dark with Alexa Fluor 488 anti-rabbit or Alexa Fluor 647 anti-mouse. ProLong Gold Antifade Mountant media containing DAPI (4',6-diamidino-2-phenylindole; Fisher Scientific, P36941) for nuclei was used to mount cells/tissues prior to imaging with a Zeiss LSM 800 AiryScan confocal microscope using the 40X water immersion lens. Configuration settings were held constant across all slides imaged. Five regions of interest of the synovium were sampled per tissue

section. Quantification of CC3⁺ apoptotic synovial cells per area was performed as previously described (21), ensuring intima and subintima were in the field of view by cross-referencing to adjacent H&E sections.

Synovial cell isolation and culture

Fresh synovial tissue biopsies were immediately placed in ice-cold DMEM, minced into 1 mm³ pieces and then enzymatically dissociated with 400 µg/mL LiberaseTM (Sigma-Aldrich, 5401127001) and 400 µg/mL DNase I (Qiagen, 79254) with gentle rotation in an incubator at 37°C for 30 minutes. Enzyme digestion was stopped with 100 mM EDTA. The cell suspension was passed through a 100 µm cell strainer into a new tube, centrifuged at 300 x g for 10 minutes at 4°C, and resuspended in cold synovial culture media (DMEM, 10% FBS, 100 U/mL penicillin, 100 µg/mL streptomycin). Cells were counted, greater than 90% viability ensured by trypan blue exclusion, and 10⁵ mixed synovial cells were plated onto glass coverslips in a 24-well plate in complete media containing RPMI 1640 (Fisher Scientific, 11-875-093), 10% FBS, 1% Pen/Strep, and 50 ng/mL macrophage colony stimulating factor (M-CSF; PeproTech, 300-25-10ug) to sustain synovial macrophages. Efferocytosis assays (below) were performed after culturing the mixed synovial cell cultures for 48 hours at 37°C and 5% CO₂. In parallel cultures, we tested the effects of pre-treatment for 48 hours with granulocyte macrophage-colony stimulating factor (GM-CSF; PeproTech, 300-03-5ug) at 0, 1, 10, and 100 ng/µL, or 50 ng/mL M-CSF + interleukin-4 (IL-4; PeproTech 400-04-5ug) at 20 ng/mL, on synovial macrophage-mediated efferocytosis.

Peripheral blood-derived macrophage cultures

Peripheral blood samples were diluted with an equal volume of 2% FBS and PBS and passed through Ficoll-Paque 1.073 density gradient medium (GE Healthcare, 17144002) using Sepmate tubes (Stemcell Technologies, 85415) by centrifugation at 1200 x g for 10 minutes at room temperature. The top layer containing enriched mononuclear cells were removed, washed twice with 1X PBS and 2% FBS/centrifuged at 300 x g for 8 minutes, twice more with complete RPMI 1640, 10% FBS, and 1% Pen/Strep, and counted. Fifty thousand mononuclear cells were plated per well in 500 μ L of complete media supplemented with 50 ng/mL M-CSF (PeproTech, 300-25-10ug), incubated at 37°C and 5% CO₂. After two days, 500 uL of complete media plus M-CSF was added. After 5-7 days, phase contrast microscopy was used to confirm the presence of morphological characteristics of macrophage differentiation (22). Non-adherent lymphocytes and platelets were removed during media changes on day 7, providing a highly enriched monolayer of blood-derived macrophages for subsequent experiments.

Blood-derived macrophage stimulation with synovial fluid

Blood-derived macrophages were stimulated with synovial fluid from patients and volunteers to model the effects of different knee OA and control conditions, respectively. Primary blood-derived macrophages from a healthy, 30-year-old female volunteer were used to allow us to model healthy human macrophages that are recruited to the synovium during OA more closely than a cell line. Macrophages were stimulated for 6 and 16 hours with synovial fluid isolated from 4 experimental groups as follows: i) healthy knees (control), ii) early-stage knee OA and high disease activity, iii) early-stage knee OA and low disease activity, and iv) late-stage knee OA and high disease activity. We defined early-stage OA as KL grades 0-2 and late-stage OA as grades 3-4. High disease activity was defined as at least moderate pain in the target

knee (KOOS pain subscale questions P2-P9 sum >7), and ‘moderate’ or ‘severe’ synovitis (grades 2-3) on OMERACT US grading in the lateral suprapatellar window. Low disease activity was defined as mild or no pain (KOOS pain subscale questions P2-P9 sum 0-7), and ‘none’ or ‘mild’ US synovitis (grades 0-1). Pilot experiments with macrophages exposed to OA synovial fluid dilutions at 50%, 25%, 12.5%, 10%, 5%, and 2.5% dissolved in 500 uL of complete RPMI 1640 media supplemented with 50 ng/mL M-CSF revealed that rounding of macrophage cell bodies (suggesting macrophage activation/response) occurred with synovial fluid as low as 5%, which was used to conduct stimulation and efferocytosis experiments.

Synovial fluid cytokine analysis

Synovial fluid from each experimental group (healthy knees, early OA in high disease activity, early OA in low disease activity, and late OA in high disease activity) without blood contamination underwent multiplex cytokine analysis (Human-focused 15-plex) using Luminex (Millipore, USA), at Eve Technologies (Calgary, Alberta) to measure pro-inflammatory cytokines such as monocyte chemoattractant protein-1 (MCP-1), IL-6, and IL-8. Samples were pre-treated with hyaluronidase (200 IU/ml) immediately prior to analysis to improve Luminex bead recovery.

Efferocytosis assay

Efferocytic efficiency and index were measured in i) synovial macrophages isolated from the synovial tissues of knee OA patients and healthy knees (control), and ii) in peripheral blood-derived macrophages, according to our published protocols (23). Jurkat cells were cultured in media containing serum-free RPMI 1640 and 2.14 μ M staurosporine (Cayman Chemical

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Company, 81590) for 24 hours at 37°C and 5% CO₂ to render cells apoptotic, prior to “feeding” to macrophages for efferocytosis assays. To differentiate engulfed vs. non-engulfed apoptotic Jurkat cell bodies, we dual-labelled apoptotic Jurkat cells with 0.005 mg of N-hydroxysuccinimido-biotin (NHS-biotin, Fisher Scientific, PI21336) dissolved in DMSO (dimethyl sulfoxide), and a cell-tracking dye (Fisher Scientific, C2927) that labels the cytosol of apoptotic cells (diluted 1:1000) and incubated at room temperature for 20 minutes in the dark. NHS-biotin was quenched for 5 minutes with an equal volume of RPMI 1640 and 10% FBS. Labelled apoptotic Jurkat cells were pelleted by centrifugation and resuspended in RPMI 1640 and 10% FBS at a concentration of 5,000,000 cells/mL. After adding 500,000 apoptotic Jurkat cells per well of macrophages, the plate was centrifuged at 200 x g for 1 minute to force contact with macrophages, then incubated for 90 minutes at 37°C and 5% CO₂. Cultures were then washed twice with PBS to stop efferocytosis and remove non-efferocytosed apoptotic cells. Macrophage-bound, but non-efferocytosed, biotinylated Jurkat cell fragments were labelled with Alexa Fluor 647-conjugated streptavidin (1:1000, Fisher Scientific, S21374) for 20 minutes in the dark along with nuclear staining using Hoechst 33342 (1:1000, Fisher Scientific). Fixation using 4% PFA (Sigma-Aldrich) in PBS for 5 minutes at room temperature was performed prior to immunolabeling macrophages for CD14 and CD68, mounting, and imaging using a Leica DMI6000 B widefield microscope and 63x oil immersion lens.

Analysis: Macrophages were identified in mixed synovial cell cultures by the presence of CD14⁺ or CD68⁺ immunofluorescence. Configuration settings were held constant across all samples imaged. Fifty macrophages were evaluated per experimental replicate. Uninternalized apoptotic cell material (streptavidin) was differentiated from internalized apoptotic material (efferosomes) (cell-tracker dye). Efferocytic efficiency was calculated as the fraction of

macrophages with at least one efferosome, and efferocytic index was determined by the average number of discrete efferosomes per macrophage.

Statistical analysis

Descriptive statistics for patient groups are presented as sample size (n), percentages, means with standard deviation (SD) and range, as appropriate. We calculated nominal p-values using t tests for between group mean differences for demographics and radiographic and histopathology grades. All experimental measurements are shown as means with 95% confidence intervals (95%CI). Shapiro-Wilk tests and Q-Q plots for normal distributions were conducted for all two-group comparisons. Apoptotic cell burden and efferocytic efficiency and index measurements in synovial tissue-derived and peripheral blood-derived macrophages were compared using unpaired, two-tailed Student's t-tests or Mann-Whitney U-tests as appropriate. Efferocytic efficiency and index data from synovial tissue-derived macrophages treated with GM-CSF, IL-4, and from peripheral blood-derived macrophages stimulated with synovial fluid, were analyzed using repeated measures mixed-effects modelling due to unequal sample sizes. Pro-inflammatory cytokine analyses from synovial fluid were analyzed using a one-way ANOVA. Dunnett's multiple testing correction was applied to all models and an alpha error of less than 0.05 was considered statistically significant. Analyses were conducted using GraphPad Prism version 9 (GraphPad Software, San Diego, CA, USA).

Results

Apoptotic cell burden in synovial tissues

Compared to volunteers with healthy knees (n = 5), patients with late-stage knee OA (n = 19) were older (p=0.0003), had higher BMI (p=0.013), and higher scores for all synovial histopathology grades (Table 1; p< 0.0001). OA synovium demonstrated increased synovial lining thickness and subintimal infiltrate, compared to healthy controls (Table 1, Figure 1A). Late-stage knee OA patients demonstrated a marked qualitative (Figure 1B) and quantitative (Figure 1C) increase in apoptotic cell burden in the synovium overall (48.4%, 95%CI 39.0-57.8) compared to healthy individuals (3.0%, 95%CI -1.4-7.5). Similar findings were obtained in the synovial intima and subintima, with a greater proportion of apoptotic cells in the intima (Figures 1D,E).

Table 1 here

Figure 1 here

Efferocytosis in synovial tissue-derived, and blood-derived macrophages

Compared to volunteers with healthy knees (n = 5), patients with late-stage knee OA (n = 19) had higher BMI (p<0.0001), radiographic severity (p<0.0001), and measures of pain (Table 2; p< 0.0001). We confirmed that nearly all CD14+ cells from OA synovial tissues co-express CD68 (Suppl. Fig. 1). The percentage of CD14+ synovial macrophages that engulfed at least one apoptotic cell (efferocytic efficiency) and the amount of efferocytosed apoptotic material per macrophage (efferocytic index) was markedly reduced in late-stage knee OA patients (eff.: 12.53%, 95%CI 9.62-15.43; index: 0.16, 95%CI 0.11-0.21) compared to healthy knees (eff: 74.60%, 95%CI 68.42-80.78; index: 0.98, 95%CI 0.71-1.23) (Figure 2A). We also measured efferocytic function in peripheral blood-derived macrophages from patients with early- and late-stage knee OA to determine if efferocytosis is also severely impaired in peripheral blood or

mainly at the local joint environment. Donors with late-stage OA had higher BMI ($p=0.0097$) and radiographic scores (Table 2, $p<0.0001$). Peripheral blood-derived macrophage-mediated efferocytic efficiency (early OA: 77.33%, 95%CI 69.74-84.92; late OA: 78.33%, 95%CI 74.05-82.62) and index (early OA: 1.05, 95%CI 0.85-1.24; late OA: 1.05, 95%CI 0.96-1.13) were equally and highly functional in patients with early- and late-stage knee OA (Figure 2B), and similar to levels seen in synovial tissue-derived macrophages from healthy knees (Figure 2A).

Unlike mice exposed to second-hand smoke (24), treatment of synovial tissue macrophages with up to 100 ng/ μ L GM-CSF had minimal to no effect on efferocytic efficiency and index (Figure 2C). In contrast, treatment of synovial tissue macrophages with 50 ng/mL M-CSF + 20 ng/mL IL-4 demonstrated nearly complete rescue of efferocytosis function (M-CSF+IL-4 eff.: 74.53%, 95%CI 69.14-79.92; index: 0.95, 95%CI 0.77-1.14) compared to M-CSF-only treatment controls (vehicle eff.: 11.03%, 95%CI 4.91-17.16; index: 0.13, 95%CI 0.06-0.21), (Figure 2D). Synovial tissue-derived macrophages from healthy knees had a more elongated morphology with numerous efferosomes and minimal non-engulfed extra-cellular apoptotic Jurkat cell fragments, whereas OA synovial macrophages had a more rounded morphology with few or no efferosomes and more non-engulfed apoptotic Jurkat cell material (Figure 2E).

Table 2 here

Figure 2 here

Synovial fluid effects on peripheral blood-derived macrophage-mediated efferocytosis

We investigated the effects of the local OA joint environment (synovial fluid) on macrophage-mediated efferocytosis *in vitro* with healthy blood-derived macrophages.

Demographics and radiographic data describing the patients who provided synovial fluid samples are presented in Table 3. No patients with late-stage OA met our criteria for low disease activity due to pain levels. Stimulation of peripheral blood-derived macrophages with 5% synovial fluid from any condition for up to 16 hours did not induce macrophage apoptosis (Figure 3A). Compared to healthy knee synovial fluid (50.88%, 95%CI 23.31-78.46%), synovial fluid from patients with early-stage OA in high disease activity caused a sustained suppression of macrophage-mediated efferocytic efficiency after 6 hours (22.25%, 95%CI 10.03-34.46%) and 16 hours (18.44%, 95%CI 10.04-26.85) of stimulation (Figure 3B). Interestingly, late-stage OA synovial fluid caused suppression of efferocytic efficiency after 16 hours (41.33%, 95%CI 26.27-56.40), whereas early-stage OA in low disease activity caused a transient suppression at 6 hours (22.67%, 95%CI 7.79-37.55%), with partial improvement after 16 hours of stimulation (46.67%, 95%CI 30.87-62.47). The amount of efferocytosed material per macrophage (efferocytic index) was impaired the most by synovial fluid from patients with early-stage OA in high disease activity, regardless of duration of exposure. Efferocytic index was significantly impaired by synovial fluid from patients with early-stage OA low activity at 6 hours but improved at 16 hours. Synovial fluid from patients with late-stage OA had minimal effect on efferocytic index (Figure 3C). MCP-1 and IL-8 were higher in synovial fluid from patients with early OA high disease activity (204.0 pg/mL, 95%CI 146.8-261.2) and late OA high disease activity (241.3 pg/mL, 95%CI 162.4-320.2) compared to healthy individuals (99.0 pg/mL, 95%CI 55.8-142.2). IL-6 was higher in patients with late-stage OA high disease activity (146.9 pg/mL, 95%CI 12.6-281.1) compared to healthy individuals (0.51 pg/mL, 95%CI 0-1.04) (Figure 3D).

Table 3 here

Figure 3 here

Discussion

Non-resolving knee inflammation contributes to poor joint health (25,26), but the underlying mechanisms are not well-understood. Synovial lining barrier macrophages are lost in mouse models of arthritis (27), and apoptotic cells should be rapidly cleared by phagocytes to maintain homeostasis (28). Thus, we were surprised to find a high proportion of CC3⁺ apoptotic cells in the synovium of patients with late-stage knee OA. CC3 is an executor protease and is unlikely to have functions in OA synovial cells that are unrelated to apoptosis. This suggested to us that macrophage-mediated clearance of apoptotic cells is impaired in OA and that their presence could contribute to the perpetuation of chronic inflammation in OA (29). In early apoptosis, factors such as lactoferrin are released to avoid recruiting inflammatory cells (30). If not removed efficiently, secondary necrosis results in membrane integrity loss and leakage of intracellular oxidative molecules and damage-associated molecular patterns (DAMPs), such as DNA, uric acid, and heat shock protein-90 (3,31,32). DAMPs are increased in OA joints, leading to the activation of the innate immune system (33–35) and production of pro-inflammatory cytokines such as TNF α and IL-1 β (25).

Since macrophages are the most abundant immune cell in OA synovium (36), we investigated whether OA synovial macrophage-mediated efferocytosis may be defective. We examined the function of synovial macrophages in mixed synovial cell cultures to recapitulate the mixed cell-cell interactions occurring *in vivo*. We found that synovial macrophages demonstrated severe impairment of efferocytosis in all OA patient samples compared to healthy synovial tissues, suggesting that impaired efferocytosis is a common feature of late-stage knee

OA. Whether efferocytosis impairment occurs universally in all stages and presentations of OA should be investigated in future studies. Nonetheless, our data indicate that severe synovial macrophage dysfunction occurs at least in patients with late-stage knee OA.

Efferocytosis is a highly conserved system with multiple redundant mechanisms, underscoring its critical role in maintaining homeostasis (37,38). Numerous macrophage receptors respond to ‘find me’ (e.g., P2Y purinergic receptors) and ‘eat me’ signals (e.g., TAM (Tyro3, Axl, Mer tyrosine kinase; MER-TK, CX3CR1, TIMs, Stabilin, and CD300 receptors) (37,39). Impaired efferocytosis therefore suggests that OA causes a complex, multi-mechanistic impairment of synovial macrophage function. Beyond clearing apoptotic, DAMP-releasing “zombie” cells, efferocytosis also triggers the release of anti-inflammatory factors such as IL-10 and TGF- β by phagocytes to promote inflammation resolution (40). This may partially explain why mesenchymal stem cell (MSC) injections reduce inflammation in mouse models of OA, as MSCs are cleared by synovial macrophages (41). Impaired efferocytosis is associated with worse disease activity in patients with systemic lupus erythematosus (42). In context with these and other studies, our findings suggest that synovial tissue macrophage dysfunction may be a key mechanism underlying the failure to resolve inflammation in knee OA.

Although efferocytosis was markedly impaired in macrophages from synovial tissues, peripheral blood-derived macrophages from patients with early- and late-stage knee OA demonstrated normal efferocytic function. Since we also found that OA synovial fluid impaired efferocytosis in healthy blood-derived macrophages, impaired macrophage-mediated efferocytosis in OA synovium most likely occurs after macrophages arrive in the joint environment, rather than in the systemic circulation. The mechanisms leading to the impairment of synovial macrophage-mediated efferocytosis however, are not clear. Interestingly, we also

found that impaired efferocytosis in synovial macrophages is reversible *ex vivo* using anti-inflammatory treatment with IL-4, which has pro-resolving effects on monocytes and macrophages (45). Although GM-CSF has been shown to rescue efferocytosis in pulmonary macrophages exposed to second-hand smoke in mice (24), it did not rescue efferocytosis impairment in OA synovial macrophages. These data suggest that efferocytosis may be reversible in OA with anti-inflammatory cytokines such as IL-4. In general, pro-inflammatory macrophages are less effective at efferocytosis than anti-inflammatory macrophages (47). Further, since pro-inflammatory cytokines and chemokines including IL-8, IL-6, and MCP-1 were increased in synovial fluid from patients with high disease activity (high pain and inflammation), these data suggest that a pro-inflammatory OA joint environment may contribute to impaired efferocytosis in synovial macrophages.

Our study has limitations. We used CD14 or CD68 expression to identify macrophages in mixed synovial cell cultures. Although these markers typically represent related but not identical macrophage sub-populations, both markers were co-expressed on nearly all OA synovial cells expressing either marker (Suppl. Fig. 1), as has been shown in other OA studies (46). Nonetheless, we cannot assume that these sub-populations are identical, nor that marker co-expression would be the same in other OA/arthritis contexts. We are also not able to rule out an effect of enzymatic digestion on synovial cell membrane receptor function and expression, although the same effect would occur in all experimental conditions. Although we did not assess the presence of apoptotic synovial cells in patients with early-stage OA, macrophage-mediated efferocytosis was impaired by synovial fluid from patients with early-stage knee OA. Thus, synovial macrophage dysfunction likely occurs early enough in knee OA to have the potential for a causal role in disease progression. This should be investigated further since the cross-sectional

design of this study limits the ability to conclude that efferocytosis impairment drives OA disease activity and progression. Our study expands our understanding of the pathophysiology and significance of synovial inflammation in knee OA, including in early-stage disease where inflammation increases the risk of radiographic progression (48,49). Interestingly, synovial fluid from patients with late-stage OA induced slightly less impairment in efferocytosis than early-stage OA. These results support the notion that pathophysiologic mechanisms in early and late OA stages are not identical and may need to be investigated separately.

To our knowledge, this is the first study to report the presence of a high burden of apoptotic synovial cells in patients with late-stage knee OA. Synovial macrophage-mediated efferocytosis is impaired in late-stage OA and by exposure of healthy macrophages to the OA joint environment. Poor joint outcomes may therefore be related to dysfunctional synovial macrophage-mediated homeostatic mechanisms in OA. Future studies should address macrophage-dependent and -independent mechanisms to determine whether impaired efferocytosis is a cause and/or consequence of inflammation in knee OA.

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Author contributions

LD contributed to data collection, carried out data and statistical analyses, interpretation of results, writing, and revising the manuscript. GB contributed to study design, data collection, carried out data and statistical analyses, interpretation of the results, and revising the manuscript. HP contributed to data collection and revising the manuscript. JR assisted with data collection, statistical analyses, and revising the manuscript. LG and BH contributed to conceptualization and design of the study and revising the manuscript. CTA contributed to conceptualization and design of the study, supported the study, and contributed to the analysis, writing, and revising the manuscript. All authors read and approved the final manuscript to be published.

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Figure Legends

Figure 1. Synovial tissue histopathology and apoptosis measures. (A) Representative H&E histopathology sections of synovium from healthy and OA patients (n = 5 per condition), demonstrating synovial lining cell thickening (arrowheads) and sub-intimal infiltrate (arrow). Scale bars = 50 μ m. (B) Representative images of synovial tissue from healthy (n = 5) and OA patients (n = 19) stained for the apoptotic cellular marker, cleaved caspase-3 (CC3, green) and Hoechst nuclear stain (blue), with dotted line indicating synovial intimal surface. Scale bars = 20 μ m. (C-E) Percent CC3+ cells in the synovium (total), intima, and subintima are shown. Data are presented as (mean $\pm$ 95% CI, *P< 0.05).

Figure 2. Efferocytosis measures in synovial tissue-derived and blood-derived macrophages. Efferocytic efficiency and index measurements are shown from (A) CD14+ synovial tissue-derived macrophages from volunteers with healthy knees (control; n = 5 vs. patients with late stage knee OA (n = 19), (B) CD14+ peripheral blood-derived macrophages from patients with early (n = 7) and late OA (n = 6), and (C) CD14+ synovial tissue-derived macrophages from patients with late-stage knee OA (n = 7) treated with or without GM-CSF. (D) CD68+ synovial tissue-derived macrophages from patients with late-stage knee OA treated with M-CSF alone (n = 6) or with M-CSF + IL-4 (n = 7). (E) Representative images of efferocytosis assays in macrophages isolated from synovial tissue of healthy and patients with OA stained for CD14 (green), cell tracer (engulfed apoptotic cell material; red), biotin (non-engulfed apoptotic cells; pink), and Hoechst nuclear stain (blue). Arrowheads indicate efferosomes. Scale bars = 10 μ m. Data are presented as (mean $\pm$ 95% CI, *P< 0.05).

Figure 3. Blood-derived macrophages treated with synovial fluid. Blood-derived macrophages stimulated with 5% synovial fluid from volunteers with healthy knees (control), or patients with early-stage OA with high disease activity, early-stage OA with low disease activity, and late-stage OA with high disease activity, for 6 or 16 hours and assessed by immunofluorescence microscopy. **(A)** Immunofluorescence confocal microscopy for apoptosis (cleaved caspase-3; CC3) was performed in parallel samples. Representative immunofluorescence images are shown indicating Hoechst (blue, nuclei), anti-CD68 (macrophage; red), and anti-CC3 (green). Positive control (macrophages treated with staurosporine for 2 hours to induce apoptosis) is shown. Scale bars = 20 μ m. **(B)** Efferocytic efficiency and **(C)** efferocytic index measures, after 6- and 16-hours stimulation with synovial fluid. **(D)** MCP-1, IL-6, and IL-8 cytokine analysis in synovial fluid from volunteers with healthy knees, or patients with early-stage OA with high disease activity, early-stage OA with low disease activity, and late-stage OA with high disease activity. Data are presented as (mean $\pm$ 95% CI, *P< 0.05).

Suppl. Figure 1. CD14+/CD68+ macrophages derived from OA synovium. (A)

Representative images of CD14 (green) and CD68 (red) co-stained synoviocytes derived from OA synovium (n = 4). Scale bars = 20 μ m. Nuclei are stained with Hoechst (blue). **(B)** Percent CD14+ and/or CD68+ cells are shown. Data are presented as mean $\pm$ 95% CI, *P< 0.05).

Tables

Table 1. Patient demographics, clinical, and histopathology characteristics of patient-derived knee synovial tissue used in apoptosis analyses.

Characteristic	Healthy knee (n = 5)	Late-Stage OA (n = 19)
Sex, n (%)		
Male	4 (80.0%)	9 (47.4%)
Female	1 (20.0%)	10 (52.6%)
Age (yrs), mean (range)	52.7 (36-70)	66.6 (53-77)
BMI (kg/m²), mean (range)	26.6 (25.4, 27.8)	32.7 (22.9-44.0)
K-L grade, n (%)¹		
0	5 (100%)	0 (0)
3	0 (0)	6 (31.6%)
4	0 (0)	13 (68.4%)
Histopathology, n (%)²		
<i>Lining</i>		
0	5 (100)	8 (42.1%)
0.5-1.5	0 (0)	9 (47.4%)
2-3	0 (0)	2 (10.5%)
<i>Sub-synovial infiltrate</i>		
0	5 (100%)	1 (5.3%)
0.5-1.5	0 (0)	14 (73.7%)
2-3	0 (0)	4 (21.0%)
<i>Vascularization</i>		
0	4 (80%)	1 (5.3%)
0.5-1.5	1 (20%)	2 (10.5%)
2-3	0 (0)	16 (84.2%)
<i>Perivascular edema</i>		
0	5 (100%)	9 (47.4%)
0.5-1.5	0 (0)	9 (47.4%)
2-3	0 (0)	1 (5.3%)
<i>Fibrosis</i>		
0	5 (100%)	0 (0)
0.5-1.5	0 (0)	15 (78.9%)
2-3	0 (0)	4 (21.1 %)
<i>Fibrin</i>		
0	5 (100%)	2 (10.5%)
0.5-1.5	0 (0)	17 (89.5%)
2-3	0 (0)	0 (0)

BMI = body mass index

¹Kellgren & Lawrence grades correspond to radiographic severity (19).

²Synovial histopathology was scored in six categories and mean scores were binned into categories 0, 0.5-1.5, and 2-3 (none, mild, moderate-severe).

Table 2. Patient demographics clinical characteristics of macrophages derived from peripheral blood and knee synovial tissue for efferocytosis analyses.

Characteristic	Synovial tissue-derived macrophages		Blood-derived macrophages	
	Healthy knee (n = 5)	Late-stage OA (n = 19)	Early-stage OA (n = 7)	Late-stage OA (n = 6)
Sex, n(%)				
Male	2 (40.0%)	11 (57.9%)	3 (42.9%)	2 (33.3%)
Female	3 (60.0%)	8 (42.1%)	4 (57.1%)	4 (33.3%)
Age (yrs), mean (range)	66.8 (27-91)	67.7 (55-80)	70.7 (59-86)	71.2 (61-82)
BMI (kg/m²), mean (range)	21.2 (14.0-27.8)	31.4 (22.8-38.0)	26.4 (21.7-34.5)	32.9 (26.3-43.5)
K-LGrade, n (%)¹				
0	5 (100%)	0 (0)	1 (14.3%)	0 (0)
1	0 (0)	0 (0)	2 (28.6%)	0 (0)
2	0 (0)	0 (0)	4 (57.1%)	0 (0)
3	0 (0)	8 (42.1%)	0 (0)	2 (33.3%)
4	0 (0)	11 (57.9%)	0 (0)	4 (66.6%)
KOOS Pain, mean (range)	100 (100)	49.6 (11-78)	59.1 (22-88)	41.8 (0-69)

BMI = body mass index, KOOS = Knee Injury and Osteoarthritis Outcome Score
¹Kellgren & Lawrence grade corresponds to radiographic osteoarthritis severity (19)

Table 3. Patient demographics and clinical characteristics of patient-derived knee synovial fluid used to stimulate blood-derived macrophages for efferocytosis analyses.

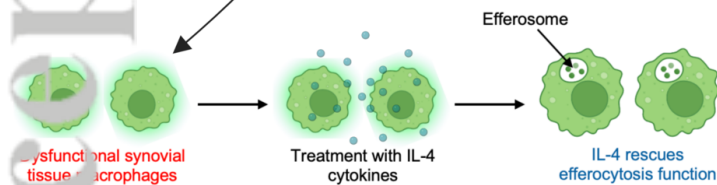
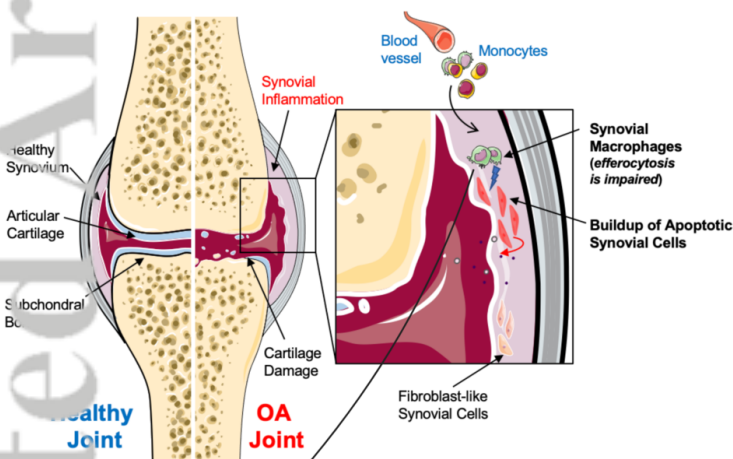
Characteristic	Healthy knees (n = 5)	Early-stage OA: High activity (n = 6)	Early-stage OA: Low activity (n = 6)	Late-stage OA: High activity (n = 6)
Sex, n(%)				
Male	2 (40.0%)	1 (16.7%)	1 (16.7%)	2 (33.3%)
Female	3 (60.0%)	5 (83.3%)	5 (83.3%)	4 (66.7%)
Age (years), mean (range)	49.6 (37-60)	64.3 (49-77)	71.0 (48-84)	59.2 (51-69)
BMI (kg/m²), mean (range)	24.5 (22.1- 28.8)	30.1 (21.9-41.7)	26.5 (22-32.5)	29.0 (22.3-42.3)
K-L Grade, n (%)¹				
0	5 (100%)	3 (50.0%)	2 (33.3%)	0 (0)
1	0 (0)	3 (50.0%)	4 (66.7%)	0 (0)
2	0 (0)	0 (0)	0 (0)	5 (83.3%)
3	0 (0)	0 (0)	0 (0)	1 (16.7%)
4	0 (0)	0 (0)	0 (0)	0 (0)
KOOS Pain, mean (range)	100 (100-100)	48.5 (19-67)	90.5 (64-97)	60.8 (58-66)
US Synovitis grade, n(%)				
0	2 (40.0%)	0 (0)	1 (16.7%)	0 (0)
1	3 (60.0%)	0 (0)	5 (83.3%)	0 (0)
2	0 (0)	4 (66.7%)	0 (0)	5 (83.3%)
3	0 (0)	2 (33.3%)	0 (0)	1 (16.7%)

BMI = body mass index, KOOS = Knee Injury and Osteoarthritis Outcome Score, US = ultrasound

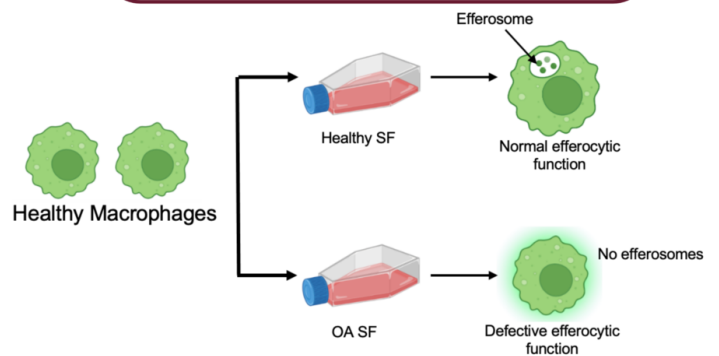
¹Kellgren & Lawrence grade corresponds to radiographic osteoarthritis severity (19).

Synovial Macrophage Dysfunction: Impaired Efferocytosis in Patients with Knee Osteoarthritis

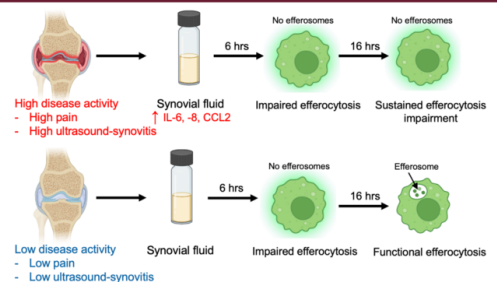
Increased apoptotic cell burden & impaired efferocytosis in late-stage OA synovial tissue



OA synovial fluid impairs efferocytosis in peripheral blood-derived macrophages



Synovial fluid from early stage OA with high disease activity sustains impaired efferocytosis

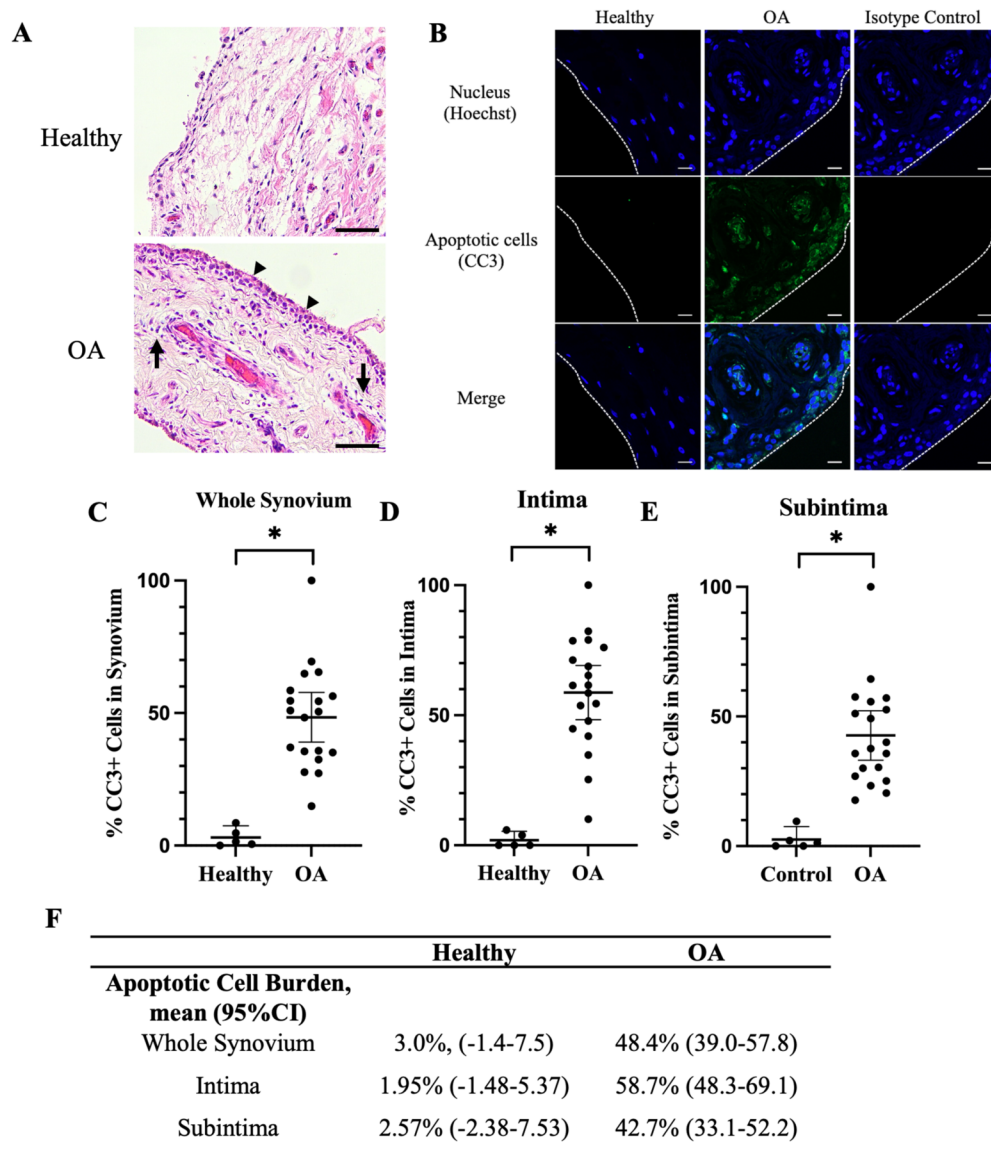


De Sordo J, Blackler G, Philpott HT, et al. Synovial macrophage dysfunction: Impaired efferocytosis in patients with knee osteoarthritis. *Arthritis Rheumatol* 2023.

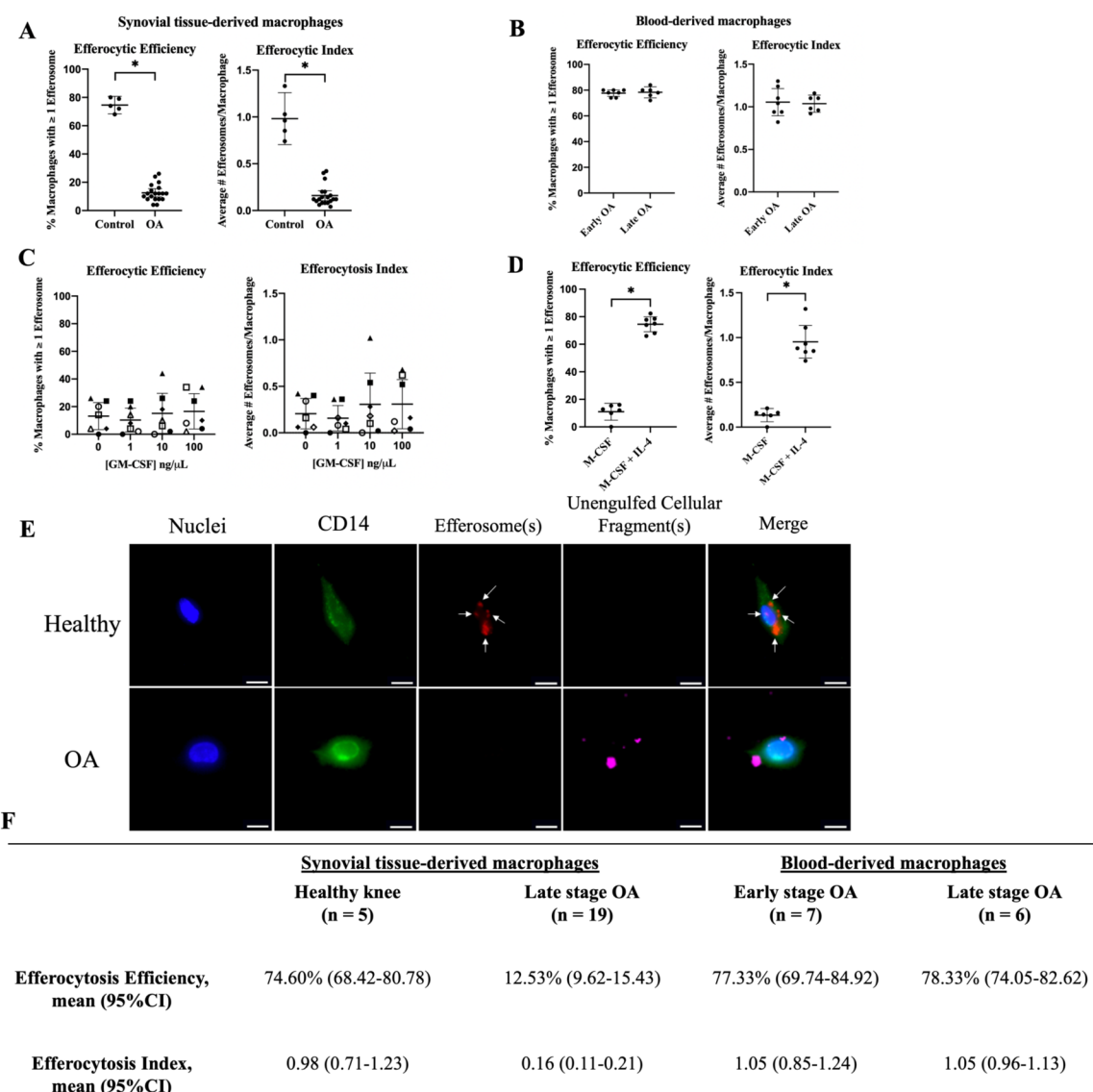
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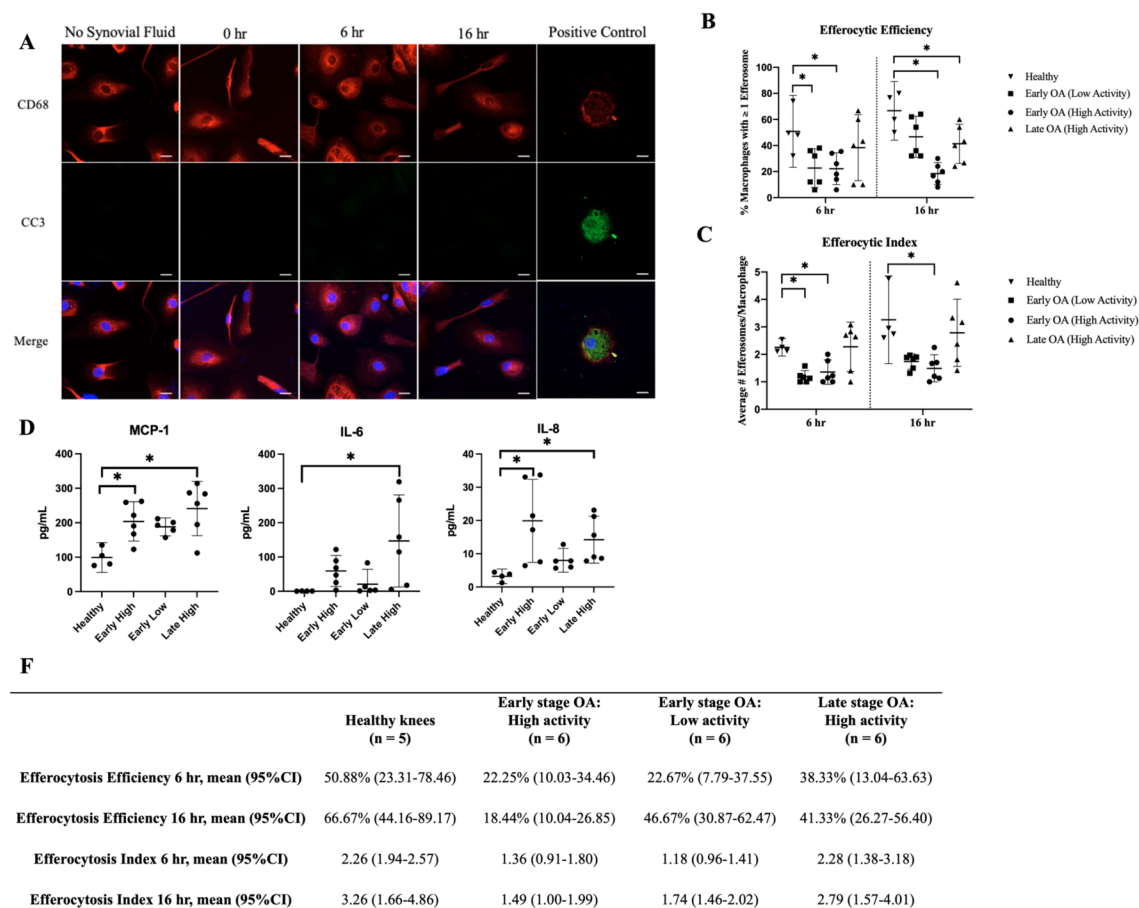
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